Predictive Maintenance Primer: Maintenance strategies.

There are three broad maintenance strategies. Reactive maintenance, also called run-to-failure, repairs equipment only after it breaks. It has no planning cost but can cause expensive unplanned downtime and secondary damage.

Preventive maintenance services equipment at fixed time or usage intervals regardless of its actual state. It reduces surprise failures but often replaces parts that still have useful life left.

Predictive maintenance, also known as condition-based maintenance, uses sensor data to estimate the actual health of the equipment and schedules work only when it is needed. The aim is to cut unplanned downtime while avoiding unnecessary replacements. The difference from preventive maintenance is that the trigger is the measured condition, not the calendar.

Predictive Maintenance Primer: Remaining useful life and the C-MAPSS data.

Remaining Useful Life (RUL) is the time left before an asset reaches a failure threshold, usually measured in operating cycles. Approaches include physics-based models, data-driven models learned from run-to-failure histories, and hybrids.

A popular benchmark is NASA's C-MAPSS turbofan engine degradation simulation. It has four subsets and 21 sensor channels. FD001 has one operating condition and one fault mode (high-pressure compressor degradation) with 100 training engines. FD002 has six operating conditions and one fault mode. FD003 has one operating condition and two fault modes (high-pressure compressor and fan degradation). FD004 has six operating conditions and two fault modes and is the hardest subset.

In the training files every engine is run until it fails. In the test files the series stops before failure and a separate file gives the true RUL at the last observed cycle.

Predictive Maintenance Primer: Evaluating RUL models.

The most common metric is the root mean squared error (RMSE) of predicted versus true RUL. Because predicting failure too late is more harmful than predicting it too early, the PHM08 challenge introduced an asymmetric scoring function. With d equal to predicted minus actual RUL, the score is exp(-d/13) - 1 when d is negative and exp(d/10) - 1 otherwise, so late predictions are penalised more heavily than early ones.

Many studies clip the training RUL target at 125 cycles, giving a piecewise linear target. In early life the sensor readings of a healthy engine look practically identical, so forcing a model to separate 200 cycles from 150 cycles left only adds noise.

A predicted RUL is only useful when it triggers an action. An alert threshold turns it into a decision, and choosing it is a trade-off: a high threshold catches more real failures but raises more false alarms, while a low threshold misses failures. The threshold should be set from the relative cost of a missed failure and an unnecessary inspection.